

Assignment 5: Deep Learning Mechanics, Attention and Variational Inferences

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1 Backpropagation Dynamics and Loss Field Gradients

The functional mechanism of network optimization relies on tracking error fields backward across internal matrices. Under a standard categorical cross-entropy loss landscape paired with a Softmax output layer, the baseline error vector $\delta^{(L)}$ simplifies strictly to:

$$\delta^{(L)} = p_{\text{pred}} - y_{\text{true}}$$

Where p_{pred} represents the output network probability vector and y_{true} maps the single-point ground-truth distribution target.

The sequential update vector for any intermediate mapping tier $w^{(l)}$ is governed by computing matrix outer products:

$$\frac{\partial \mathcal{L}}{\partial w^{(l)}} = \delta^{(l)} (a^{(l-1)})^T$$

When utilizing the standard Rectified Linear Unit (ReLU) function, the gradient operator vanishes identically for any non-positive initialization range ($x \leq 0$). This mathematical constraint results in the permanent inactivation of specific structural features, an optimization phenomenon known as the dying node pathology.

2 Architectural Topologies and Attention Matrices

Unlike general fully-connected operators that lack spatial orientation context, Convolutional Neural Networks (CNNs) enforce translation invariance restrictions by parameterizing localized receptive fields. This structural weight sharing reduces spatial complexity metrics significantly compared to linear networks.

In modern deep attention configurations, sequence tokens map to independent linear vector fields labeled Queries (Q), Keys (K), and Values (V). The alignment scores are scaled inversely to prevent dot-product magnitude explosions:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_{\text{features}}}} \right) V$$

3 Variational Inferential Bounding (ELBO)

In latent generative frameworks (VAEs), direct optimization of the true marginal log-likelihood $\ln p(x)$ is computationally intractable. Applying Jensen's integral inequalities allows the optimiza-

tion target to settle into the Evidence Lower Bound (ELBO):

$$\ln p_{\theta}(x) = \mathcal{L}_{\text{ELBO}}(\phi, \theta; x) + D_{\text{KL}}(q_{\phi}(z|x) \parallel p_{\theta}(z|x))$$

Since relative entropy fields satisfy non-negativity parameters ($D_{\text{KL}} \geq 0$), maximizing the functional ELBO guarantees steady improvement of the true underlying generative data distribution.